

# Synthesis Memo

Query: What are the operational integration barriers for ML in collision avoidance?

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## Operational Integration Barriers for Machine Learning in Collision Avoidance: A Research Synthesis

**Executive Summary:** While machine learning approaches show promise for spacecraft collision avoidance, multiple technical and operational barriers currently prevent their integration into operational decision-making systems. Key obstacles include insufficient reliability for safety-critical decisions, fundamental modeling challenges with non-deterministic orbital dynamics, and dataset limitations that constrain model generalizability.

### The Reliability Gap for Operational Decision-Making

The most significant barrier to ML integration is the technology's current inability to meet the stringent reliability requirements for collision avoidance operations. Research demonstrates that "AI/ML approaches investigated to date have not yet demonstrated sufficient reliability for operational collision avoidance decision making" (mashiku2025, sec6\_p1). This reliability gap is particularly critical given that collision avoidance decisions directly impact satellite safety and the broader space environment. While ML models can improve upon baseline forecasting approaches, as demonstrated through competitive benchmarking (uriot2022, sec7\_p1), the margin of improvement has not yet reached operational thresholds necessary for autonomous decision authority.

The European Space Agency's 2019 Collision Avoidance Challenge provided crucial insights into this reliability challenge by enabling "for the first time, the study of the use of ML methods in the domain of spacecraft collision avoidance" through public dataset release (uriot2022, sec7\_p1). This competition attracted 96 teams and generated 862 submissions (uriot2022, sec1.1\_p1), establishing a robust foundation for understanding ML capabilities and limitations in this domain. However, the results revealed that while ML models could exceed naive forecasting benchmarks, they faced substantial challenges in achieving the consistency required for operational deployment.

### Fundamental Modeling Challenges

Beyond empirical performance limitations, ML integration faces deep theoretical obstacles rooted in the physics of orbital mechanics and operational realities. A critical barrier is that "predictions of future measurement updates are nearly impossible to model deterministically since they are not entirely physics-based" (mashiku2025, sec6\_p1). This fundamental unpredictability stems from the nature of tracking data acquisition, which depends on sensor availability, atmospheric conditions, and other factors that resist deterministic modeling.

Equally problematic is the difficulty of incorporating routine satellite operations into ML frameworks. The evidence indicates that "routine maneuvers as part of normal operations are difficult or impossible to model or incorporate within a model" (mashiku2025, sec6\_p1). Since satellites regularly perform planned maneuvers for orbit maintenance, constellation management, and mission objectives, any collision avoidance system must account for these interventions. The inability to reliably integrate such operational activities represents a fundamental gap between ML model assumptions and operational reality.

These modeling challenges are particularly acute for time-series prediction approaches. While research suggests that "investigating Recurring Neural Networks (RNRs) and LSTM models to ingest time-series based information could eventually provide more reliable predictions at TCA" (mashiku2025, sec6\_p1), the conditional phrasing and future tense underscore that current architectures remain inadequate. The challenge lies in developing models that can simultaneously capture orbital dynamics, incorporate observational uncertainties, and adapt to operational interventions--a combination that has proven elusive.

## **Dataset Limitations and Training Challenges**

The operational integration of ML is further constrained by inherent limitations in available training data. The competition dataset, while representing "an important historical record of risky conjunction events that occurred in LEO" (uriot2022, sec4\_p1), suffers from structural imbalances that complicate model development. Researchers encountered "several challenges, mostly derived from the unavoidable unbalanced nature of the dataset" (uriot2022, sec7\_p1), which limited the applicability of standard ML pipelines and required specialized preprocessing approaches.

This dataset imbalance reflects the operational reality that high-risk collision events are, by design and fortunate circumstance, relatively rare. However, this rarity creates a classic machine learning challenge: models must generalize from limited examples of the most critical events they are designed to predict. The competition design sought to address forecasting needs by framing the problem as predicting "the evolution of the collision risk over time" (uriot2022, sec0\_p1) from conjunction data messages received up to two days before closest approach (catulo2023, sec1\_p5). Yet even with this temporal structure, the fundamental scarcity of high-risk events constrains model training.

## **The Benchmark Challenge and Incremental Improvement**

An unexpected operational barrier emerged from the discovery that simple baseline methods perform remarkably well. Competition results showed that "naive forecasting models have surprisingly good performances and thus are established as an unavoidable benchmark for any future work in this subject" (uriot2022, sec7\_p1). This finding creates a high bar for operational adoption: ML approaches must not only work reliably but must demonstrably outperform simpler alternatives that are easier to validate, explain, and maintain.

The benchmark challenge reflects a broader pattern in safety-critical systems where operational integration requires clear performance advantages that justify increased system complexity. While the evidence confirms that "ML models can improve upon such a benchmark, hinting at the possibility of using ML to improve the decision-making process in collision avoidance systems" (uriot2022, sec7\_p1), the use of "hinting at the possibility" rather than stronger language suggests that current improvements remain modest and inconsistent.

## **Uncertainty Quantification and Explainability Requirements**

Operational collision avoidance systems must provide decision-makers with robust uncertainty quantification to support risk assessment. Current ML approaches face criticism for inadequate uncertainty characterization, with researchers noting that future systems must develop "approaches that incorporate uncertainty quantification in the predictions" (mashiku2025, sec6\_p1). Without reliable uncertainty bounds, operators cannot properly assess the confidence they should place in ML predictions, undermining trust and preventing operational integration.

Post-competition research has explored various technical approaches to address these limitations, including Bayesian deep learning methods and probabilistic programming frameworks (catulo2023, sec1\_p7). These efforts recognize that operational systems require not just point predictions but full characterizations of

prediction confidence and potential error modes. However, the ongoing nature of this research indicates that operationally adequate solutions have not yet emerged.

## Implications and Open Questions

The evidence reveals a complex integration challenge that extends beyond mere technical performance. While researchers maintain that "ML approaches will be essential in improving collision avoidance analyses and decision making processes in the near future" (acciarini2021, sec1\_p7), achieving this vision requires addressing multiple interconnected barriers simultaneously. The release of open-source tools like the Kessler library (acciarini2021, sec0\_p2) and public datasets represents important infrastructure development, yet these resources cannot overcome fundamental modeling limitations.

Critical open questions remain regarding the path to operational integration. First, what specific reliability thresholds must ML systems achieve before operators will trust them with safety-critical decisions? Second, can hybrid approaches combining physics-based models with ML augmentation circumvent some current limitations? Third, what institutional and regulatory frameworks are needed to validate ML-based collision avoidance systems? The evidence suggests that purely data-driven approaches face inherent limitations, but the optimal integration strategy remains unclear.

Additionally, the evidence indicates promising but unexplored directions, particularly "adaptive screening volumes that optimize computational resources while maintaining safety margins" (mashiku2025, sec6\_p1). Such applications might offer near-term operational value while avoiding direct responsibility for critical decisions, potentially providing a stepping-stone toward fuller integration.

In conclusion, operational ML integration for collision avoidance remains aspirational rather than imminent, constrained by reliability gaps, fundamental modeling challenges, dataset limitations, and uncertainty quantification requirements. Progress will require not just improved algorithms but potentially new problem formulations that better align ML capabilities with operational needs.

## References

- acciarini2021 -- Kessler: An open-source Python package for machine learning applied to collision avoidance (2021)
- catulo2023 -- Machine learning in collision avoidance: post-competition analysis (2023)
- mashiku2025 -- Future Directions and Conclusions: AI/ML in collision avoidance (2025)
- uriot2022 -- Spacecraft Collision Avoidance Challenge: Design, results and lessons learned (2022)